

RAR: Reconstruction-Aware Routing with Dual-Head Calibration for Encrypted Traffic Class-Incremental Learning

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Abstract

Encrypted-traffic classifiers must continuously absorb emerging applications and threats without retraining from scratch. Although class-incremental learning (CIL) enables continual model updates, existing methods typically rely on replay, knowledge distillation, or regularization to mitigate catastrophic forgetting. However, these mechanisms remain largely sample-agnostic, lacking a way to estimate whether an incoming sample is more closely associated with previously acquired or newly emerging knowledge.

We observe that an autoencoder trained on frozen encoder features of replayed old-class samples reconstructs old samples with lower error than new ones. This observation suggests that reconstruction error reflects *knowledge ownership*, i.e., the probability that a sample belongs to old or new knowledge.

Based on this observation, we propose Reconstruction-Aware Routing (RAR), which converts reconstruction error into a sample-level knowledge ownership probability and uses it to softly weight two dedicated classification heads. A lightweight logit calibration module further aligns the dual-head outputs into a unified prediction space.

Experiments on six public encrypted-traffic datasets demonstrate that RAR reduces forgetting by 35.5%–67.5% over competitive CIL baselines while improving mixed-class accuracy by up to 15.9 percentage points. Ablation studies further validate the effectiveness of knowledge ownership estimation, dual-head soft routing, and logit calibration. These results demonstrate that sample-level knowledge ownership provides a principled basis for sample-aware knowledge transfer in encrypted-traffic class-incremental learning.

CCS Concepts

• **Security and privacy** → **Network security**; • **Computing methodologies** → **Transfer learning**; • **Networks** → **Traffic classification**.

Keywords

Encrypted Traffic Classification, Class-Incremental Learning, Knowledge Distillation, Reconstruction-Aware Routing

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1 Introduction

Encrypted-traffic classifiers identify applications and threats from packet metadata, length distributions, and timing patterns alone, and the set of classes they must recognize grows continuously. New applications launch, new malware families appear, and each addition forces the classifier to recognize a class it has never seen. Operators run these classifiers under tight latency budgets and cannot retrain from scratch on every addition; the retraining cost grows with the class history.

Class-incremental learning (CIL) targets this need by adding classes without retraining on the full history. Existing CIL methods, including LwF [3], iCaRL [5], BiC [6], and DER [7], mitigate forgetting through two mechanisms: replaying a small buffer of historical samples and distilling the old model into the new one. Applied to encrypted traffic, three structural gaps remain open. First, expanding the classification head makes old and new classes compete for one shared decision boundary. Second, uniform distillation treats every sample identically, regardless of whether it carries old or new knowledge. Third, replay depends on a limited buffer that cannot represent the full old-class distribution.

We observe that a module reconstructed from the old model and the replay buffer tends to reproduce old-class samples with lower reconstruction error and new-class samples with higher reconstruction error. Reconstruction error therefore yields, for each input, the probability that the input carries old versus new knowledge. We call this per-input signal *knowledge ownership estimation*.

RAR (Reconstruction-Aware Routing) turns the knowledge ownership probability into a soft routing weight over two classification heads: an old-knowledge head trained only on old classes and a new-knowledge head trained only on new classes. Both heads contribute to every prediction, weighted by the sample's estimated old/new knowledge probability, and a logit calibration layer unifies their outputs into a joint prediction over all classes. A reconstruction temperature τ controls routing softness: small τ approaches deterministic head selection; large τ smoothly fuses both heads. The design follows a single line: knowledge ownership estimation, soft routing, dual knowledge paths, calibration.

We make four contributions.

- (1) **Sample-level knowledge ownership estimation.** We estimate each sample's old/new knowledge probability from reconstruction error, enabling sample-aware knowledge balancing beyond sample-agnostic knowledge transfer.
- (2) **Dual-head knowledge path decoupling.** We route old and new knowledge paths to separate heads, removing the shared-classifier competition that causes forgetting.
- (3) **Logit calibration over dual knowledge paths.** We calibrate the two heads' outputs into a unified classification

space, raising mixed-class accuracy by 2.3 percentage points (from 87.82% to 90.12%) over an uncalibrated dual-head baseline.

- (4) **Validation across six encrypted-traffic datasets.** RAR reduces forgetting by 35.5%–67.5% and improves mixed-class accuracy by up to 15.9% over the strongest classical CIL baseline (Adapter) on six public datasets (NUDT-Mobile, CIC-DoHBrw, CIC-IoT2023, CSTNET-TLS1.3, ISCX-Tor, EBSNN-WebApp), and the reduction persists across nine traffic-classification backbones.

The remainder of this paper evaluates the contribution. §2 formalizes the encrypted-traffic CIL problem and its three structural gaps. §3 presents RAR’s mechanism, from reconstruction error through soft routing to calibration, and closes with a training algorithm (Algorithm 1). §4 reports empirical evidence on six datasets: RAR against classical CIL strategies on ET-BERT, a backbone study confirming that naive forgetting persists across nine encoders, a component ablation, and robustness under multi-step increments and tight replay budgets. §5 positions RAR against prior work. §6 summarizes the contributions, limitations, and future directions.

2 Preliminaries

2.1 The Encrypted-Traffic CIL Setup

We formalize the problem as class-incremental learning (CIL). At the base step, we train on K_0 classes. At each incremental step, K_i new classes arrive with a small replay buffer of old samples.

At the incremental step T_i the classifier must satisfy two competing objectives. It must acquire the new classes with high accuracy, and it must retain the old classes despite the head-expansion operation that adds K_i output dimensions to the classification layer. The replay buffer typically holds a fixed small number of samples per old class, which is the only bridge between the base step and the incremental step.

Encrypted-traffic classification adds three domain-specific pressures on top of generic CIL. Packet payloads are encrypted and uninformative; the classifier must rely on packet length, timing, direction, and TLS handshake features. Class boundaries shift as new applications launch and new malware families appear. And the deployed classifier must run online under tight latency budgets, so retraining on the full history on every addition is rarely practical.

2.2 The ET-BERT Backbone

RAR adopts ET-BERT [4] as its frozen encoder. ET-BERT is a Transformer pre-trained on large-scale unlabeled encrypted traffic through two self-supervised tasks: a Masked BURST Model that captures byte-level correlations within transmission bursts, and a Same-origin BURST Prediction that models sequential relationships across consecutive bursts. The pre-trained tokenizer converts raw packet sequences into fixed-length input tokens, and the encoder produces class-separable features that transfer across encrypted-traffic domains without per-task labels. We freeze the encoder at its pre-trained weights and build the routing mechanism on top of its representations; the frozen encoder is shared by both classification heads and the reconstruction autoencoder.

2.3 Three Structural Gaps

Three structural gaps in prior CIL methods explain why forgetting persists on encrypted-traffic classification; closing any one of them in isolation is insufficient.

Shared classifier competition. Expanding the classification head to accommodate new classes forces old and new outputs to share one decision boundary [5, 6]. The shared cross-entropy loss produces gradients that point in different directions for the two groups, and the boundary drifts toward whichever group has larger or more recent gradients. The drift is training-induced forgetting and is independent of the replay buffer size.

Knowledge-unaware distillation. Distillation-based CIL methods (LwF [3], iCaRL [5]) apply the same distillation loss to every sample, regardless of whether the sample carries old or new knowledge. The uniform signal cannot tell a sample that is close to the old-knowledge manifold from one that is far from it; both contribute the same gradient to the new model. Distillation alone therefore preserves the average old-knowledge behavior but loses the per-sample distinction that incremental classification needs.

Limited replay representation. Replay-based CIL methods (DER [7], iCaRL [5]) depend on a small buffer of stored old samples that cannot represent the full old-class distribution. Below a buffer-size threshold the replay buffer stores only a few local modes of each old class, and the autoencoder or replayed features that depend on the buffer inherit the under-representation. The forgetting rate therefore degrades sharply below a buffer-size threshold that the deployment context often cannot meet.

3 Method

RAR’s design follows a single line: knowledge ownership estimation, soft routing, dual knowledge paths, calibration. A reconstruction autoencoder over frozen-encoder features produces a per-sample old-versus-new knowledge probability. RAR uses that probability as a soft routing weight over two classification heads: a frozen old-class head inherited from the base step, and a trainable new-class head equipped with a lightweight adapter. A logit calibration step then maps the two heads’ outputs into a single classification space. The rest of this section defines each piece, justifies every choice against an alternative we tried or considered, and closes with the assumptions that bound the mechanism.

3.1 Problem Formulation

Let $f : \mathcal{X} \rightarrow \mathbb{R}^d$ be a frozen encoder pre-trained on large-scale encrypted traffic data (ET-BERT [4]). At the base step, we train a classification head $h_0 : \mathbb{R}^d \rightarrow \mathbb{R}^{K_0}$ on K_0 classes with a labeled dataset $\mathcal{D}_0 = \{(x_i, y_i)\}_{i=1}^{N_0}$. At an incremental step, K_{new} new classes arrive with training data $\mathcal{D}_{\text{new}}$, and a small replay buffer $\mathcal{B}$ of old-class samples is retained from $\mathcal{D}_0$. The standard incremental-learning operation expands the head to $h : \mathbb{R}^d \rightarrow \mathbb{R}^{K_0+K_{\text{new}}}$ and trains on $\mathcal{B} \cup \mathcal{D}_{\text{new}}$ with a distillation loss anchored by the base-step model. RAR replaces the expanded shared head with two independent heads: a frozen old-class head $h_{\text{old}} \equiv h_0$ and a trainable new-class head $h_{\text{new}} : \mathbb{R}^d \rightarrow \mathbb{R}^{K_{\text{new}}}$. It then routes each sample x through a

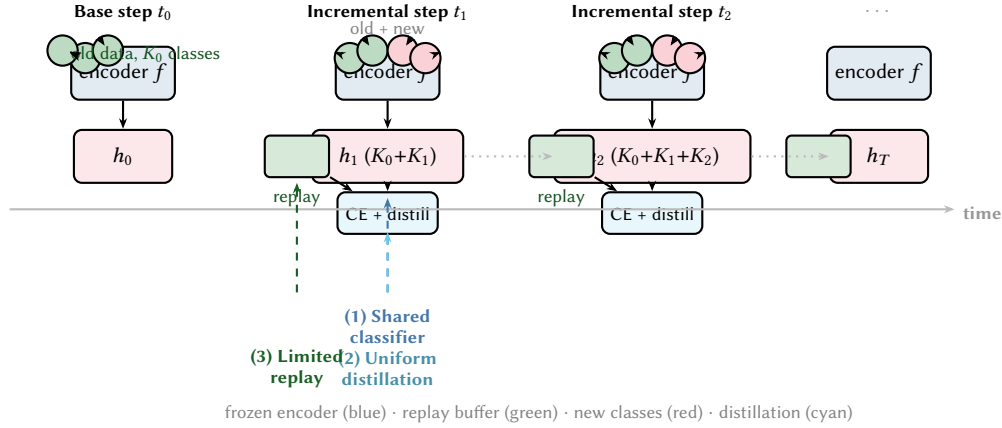


Figure 1: Class-incremental learning setup in encrypted-traffic classification.

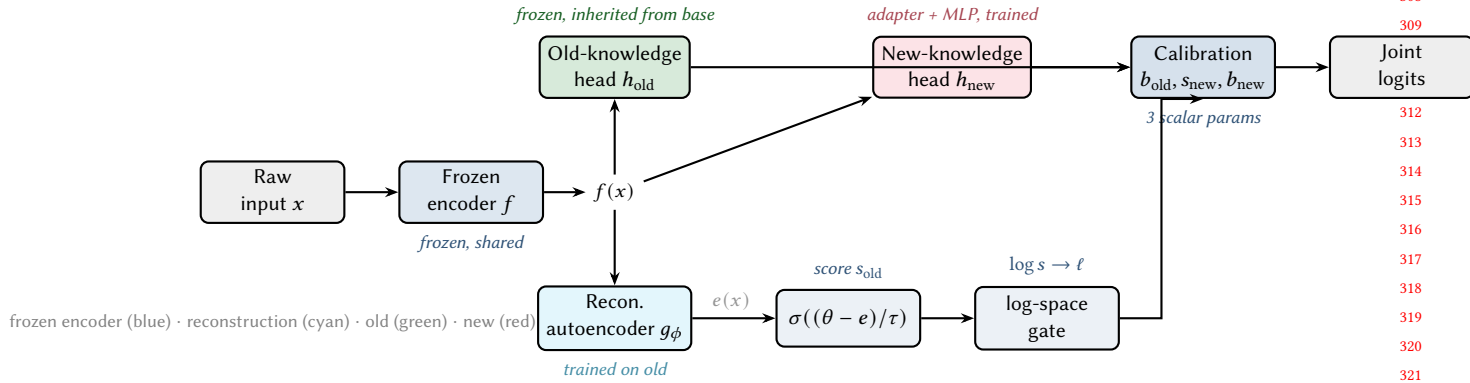


Figure 2: RAR architecture overview.

per-sample weight derived from reconstruction error. The following subsections define the reconstruction signal (§3.2), the routing mechanism (§3.3), the calibration step (§3.4), and the training procedure (§3.5).

3.2 Knowledge Ownership Estimation

RAR represents each sample’s location in the old-versus-new knowledge space by a single per-sample probability derived from reconstruction error. This probability replaces the class label or learned embedding that prior methods use to distinguish old from new samples.

We train a reconstruction autoencoder g_ϕ on frozen-encoder features of replayed old samples, so g_ϕ learns the manifold that the frozen encoder f produces for old classes. The reconstruction error therefore tends to be small on old-class samples and large on new-class samples:

$$e(x) = \frac{1}{d} \|g_\phi(f(x)) - f(x)\|_2^2. \quad (1)$$

We fit a decision threshold θ from the old and new reconstruction-error distributions (old 95th percentile vs. new 5th percentile), then

convert the error to per-sample ownership scores:

$$s_{old}(x) = \sigma\left(\frac{\theta - e(x)}{\tau}\right), \quad (2)$$

$$s_{new}(x) = 1 - s_{old}(x), \quad (3)$$

where τ controls the sharpness of the sigmoid. The fitted threshold and temperature are fixed after the pre-training stage; only $e(x)$ is computed per input at inference.

We use reconstruction error because it requires no labels and no class boundaries: it asks whether $f(x)$ lies on the manifold g_ϕ was trained to reproduce, a strictly geometric question. Distillation-based alternatives (LwF [3], iCaRL [5]) compare soft outputs of an old model and a new model and therefore inherit every class-boundary bias the old model already encoded. Replay-based alternatives (DER [7]) compare stored features and inherit the storage buffer’s bias. Reconstruction error is the only one of the three that depends on a learned manifold, which we control by training on replayed old samples only.

The empirical foundation for p_{old} is separability: across the six datasets we evaluate, the reconstruction-error distribution separates old from new with mean AUC 0.894 (Figure 4). The separability

holds on every dataset, which licenses the routing mechanism on the full benchmark.

3.3 Soft Routing and Dual Knowledge Paths

RAR routes each input through two knowledge paths, a frozen old-class head h_{old} inherited from the base step and a trainable new-class head h_{new} equipped with a lightweight adapter, weighted by the ownership scores $s_{\text{old}}(x)$ and $s_{\text{new}}(x)$ from Eqs. (2)–(3).

h_{old} is the base-step classifier, loaded from the checkpoint and frozen. h_{new} is trained on the new classes through a lightweight residual adapter and a LayerNorm+MLP pathway. Both heads share the same frozen features $f(x)$.

We apply the routing signal in log space. Let $\ell_{\text{old}}(x) \in \mathbb{R}^{K_0}$ and $\ell_{\text{new}}(x) \in \mathbb{R}^{K_{\text{new}}}$ be the raw logit vectors from the two heads. The gated logits are computed by adding the log of each head's ownership score:

$$\tilde{\ell}_{\text{old}}(x) = \ell_{\text{old}}(x) + \log s_{\text{old}}(x), \quad (4)$$

$$\tilde{\ell}_{\text{new}}(x) = \ell_{\text{new}}(x) + \log s_{\text{new}}(x). \quad (5)$$

The two gated vectors are concatenated into a joint prediction over all $K_0 + K_{\text{new}}$ classes:

$$\hat{y} = \text{softmax}([\tilde{\ell}_{\text{old}}(x); \tilde{\ell}_{\text{new}}(x)]). \quad (6)$$

In softmax space, adding $\log s$ is equivalent to multiplying the head's class probabilities by s : the old-head probabilities are scaled by $s_{\text{old}}(x)$ and the new-head probabilities by $s_{\text{new}}(x)$, so both heads contribute to every prediction in proportion to the sample's estimated old and new knowledge. As $\tau \rightarrow 0$ the sigmoid sharpens toward a step function and s_{old} approaches 0 or 1; as $\tau \rightarrow \infty$ the sigmoid flattens and both heads contribute near-equally. §4.4 reports the empirical trade-off.

We separate old and new into independent heads because head expansion, the standard incremental-learning operation, makes old and new classes compete for one shared decision boundary [5, 6]. A shared classifier trained on a replay buffer of old samples and a fresh batch of new samples minimizes the same cross-entropy loss for both groups. The resulting gradients point in different directions for each group, and the boundary drifts toward whichever group has larger or more recent gradients. Independent heads eliminate the shared-decision-boundary failure mode at the cost of a calibration step that recombines the two outputs; Table 5 reports this contribution as an ablation.

We use soft routing because hard assignment routes every sample to a single head and discards the second head's view entirely. On samples near the reconstruction-error decision boundary, the discarded head often holds the correct prediction. Soft routing keeps both views available and shifts their contribution with the input, so the calibration step is structurally necessary: the two heads cannot be averaged directly because they output logits over disjoint class subsets.

3.4 Logit Calibration

Each head outputs logits over its own class subset; a calibration layer removes the scale mismatch between the two paths before their logits are combined.

The two heads live on different class axes and may produce logits at different magnitudes. A naive concatenation would let the head with larger logits dominate Eq. (6) regardless of the routing scores. Calibration equalizes the two heads' output scales on a held-out validation set before the logits are combined.

We learn three scalar parameters: a subtractive bias $b_{\text{old}} \in \mathbb{R}$ on the old-head logits, a multiplicative scale $s_{\text{new}} \in \mathbb{R}^+$, and an additive bias $b_{\text{new}} \in \mathbb{R}$ on the new-head logits:

$$\ell'_{\text{old}} = \tilde{\ell}_{\text{old}} - b_{\text{old}}, \quad (7)$$

$$\ell'_{\text{new}} = s_{\text{new}} \cdot \tilde{\ell}_{\text{new}} + b_{\text{new}}, \quad (8)$$

where $\tilde{\ell}_{\text{old}}, \tilde{\ell}_{\text{new}}$ are the gated logits from Eqs. (4)–(5). The concatenation in Eq. (6) then uses $[\ell'_{\text{old}}; \ell'_{\text{new}}]$. We fit $(b_{\text{old}}, s_{\text{new}}, b_{\text{new}})$ by minimizing cross-entropy on a held-out validation set of mixed old and new samples at the end of training, with the heads frozen.

Calibration does not change which classes the heads cover; it only equalizes their output scales. The classification decision remains a weighted combination of the two heads' calibrated logits, where the weights come from the routing probability. Table 5 reports the contribution of calibration as an ablation.

3.5 Training Objective

RAR training proceeds in two stages.

Stage 1: Pre-training the reconstructor. We fit the reconstruction autoencoder g_ϕ on frozen-encoder features of replayed old samples by minimizing $\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{B}|} \sum_{x \in \mathcal{B}} \|g_\phi(f(x)) - f(x)\|_2^2$, then freeze g_ϕ and fit the decision threshold θ and temperature τ from the old-versus-new reconstruction-error distributions. The autoencoder receives no further gradients.

Stage 2: Joint training. We optimize a composite loss over the adapter, the new-head pathway (a LayerNorm, MLP, and new-class classifier), and a distillation term:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \alpha \mathcal{L}_{\text{distill}} + \beta \mathcal{L}_{\text{local}} + \gamma \mathcal{L}_{\text{margin}} + \delta \mathcal{L}_{\text{reg}}, \quad (9)$$

$$\mathcal{L}_{\text{distill}} = \frac{1}{Z} \sum_{x \in \mathcal{B}} s_{\text{old}}(x) D_{\text{KL}}(h_{\text{old}}^{\text{teacher}}(x) \| h_{\text{old}}(x)). \quad (10)$$

Here $\mathcal{L}_{\text{CE}}$ is cross-entropy on the concatenated output over all classes; $\mathcal{L}_{\text{local}}$ is cross-entropy restricted to the new head on new-class samples; $\mathcal{L}_{\text{margin}}$ penalizes the old head for outscoring the new head on new-class samples, weighted by $s_{\text{new}}(x)$; and $\mathcal{L}_{\text{reg}} = \|a(x) - f(x)\|_2^2$ regularizes the adapter shift from the frozen features. The frozen encoder f and the frozen old-head classifier h_{old} receive no gradients. The replay buffer is sampled at a fixed ratio (1%) of the old training data; we report the buffer-size sweep in §4.5.

Stage 3: Post-hoc calibration. After the main training stage, the three calibration parameters $(b_{\text{old}}, s_{\text{new}}, b_{\text{new}})$ in Eqs. (7)–(8) are fitted on a held-out mixed validation set with the heads frozen, by minimizing cross-entropy on the calibrated output.

Algorithm. Algorithm 1 summarizes the full training and inference procedure.

The temperature τ is the only knob we expose at inference; it trades head determinism against head fusion. We tune τ on a held-out validation set per dataset and report the sensitivity in §4.4. The knob exists because different deployments prefer different

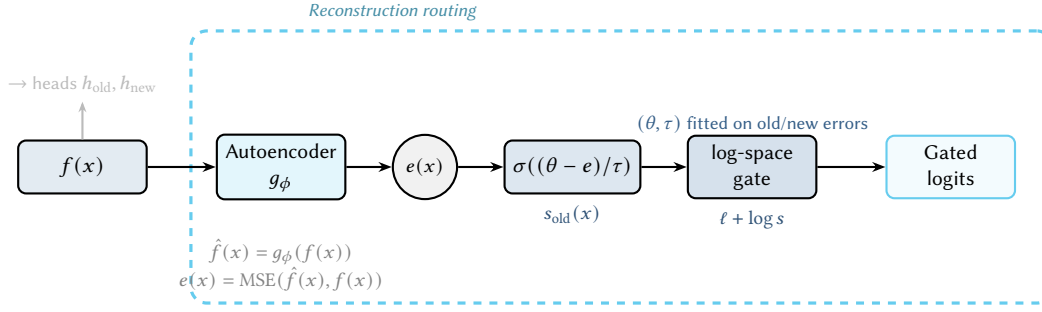


Figure 3: Reconstruction routing mechanism.

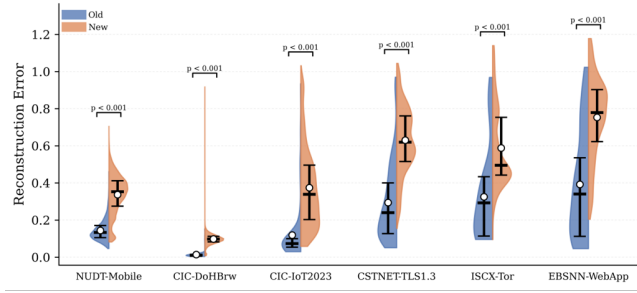


Figure 4: Per-dataset reconstruction-error distributions (old vs. new).

operating points: a deployment with strict latency budgets can lower τ toward determinism and recover a single-head equivalent; a deployment with mixed traffic prefers the mid range.

3.6 Applicability and Assumptions

RAR is designed under two key assumptions that define its applicability.

First, replay memory should provide representative coverage of previously learned knowledge. If replay samples fail to capture the old-class distribution, the reconstructor may learn an incomplete representation of old knowledge, leading to inaccurate ownership estimation. §4.5 evaluates different replay buffer sizes and shows that RAR remains robust even under a memory ratio of 0.005.

Second, reconstruction error should exhibit sufficient separability between old and emerging knowledge. When old and new classes have highly similar feature representations, reconstruction errors may overlap, reducing the reliability of knowledge ownership estimation. Figure 4 confirms statistically significant separation across all six datasets ($p < 0.001$).

These assumptions characterize the applicability of RAR rather than strict limitations. RAR is most effective when replay samples preserve representative old knowledge and when old and emerging knowledge exhibit distinguishable representation patterns.

Algorithm 1: Reconstruction-Aware Routing (RAR)

Input: Frozen encoder f ; replay buffer $\mathcal{B}$; new-class stream $\mathcal{D}_{\text{new}}$; reconstructor g_ϕ ; temperature T
Output: Incremental classifier F

- 1 **for each batch from $\mathcal{B} \cup \mathcal{D}_{\text{new}}$ do**
- 2 **Feature extraction;**
- 3 $z \leftarrow f(x)$ (frozen, no gradient);
- 4 **Reconstruction error;**
- 5 $e(x) = \frac{1}{d} \|g_\phi(z) - z\|_2^2$;
- 6 **Ownership probability;**
- 7 $s_{\text{old}} = \sigma\left(\frac{\theta - e(x)}{\tau}\right)$;
- 8 $s_{\text{new}} = 1 - s_{\text{old}}$;
- 9 **Dual-head routing (log-space);**
- 10 $\tilde{\ell}_{\text{old}} = \ell_{\text{old}} + \log s_{\text{old}}$;
- 11 $\tilde{\ell}_{\text{new}} = \ell_{\text{new}} + \log s_{\text{new}}$;
- 12 $\hat{y} = \text{softmax}([\tilde{\ell}_{\text{old}}; \tilde{\ell}_{\text{new}}])$;
- 13 **Training objective;**
- 14 $\mathcal{L} = \mathcal{L}_{\text{CE}} + \alpha \mathcal{L}_{\text{distill}} + \beta \mathcal{L}_{\text{local}} + \gamma \mathcal{L}_{\text{margin}} + \delta \mathcal{L}_{\text{reg}}$;
- 15 **end**
- 16 **Post-hoc calibration;**
- 17 $\ell'_{\text{old}} = \tilde{\ell}_{\text{old}} - b_{\text{old}}$;
- 18 $\ell'_{\text{new}} = s_{\text{new}} \cdot \tilde{\ell}_{\text{new}} + b_{\text{new}}$;
- 19 Fit $(b_{\text{old}}, s_{\text{new}}, b_{\text{new}})$ on the validation set (heads frozen);
- 20 **return F**

4 Experiments

4.1 Experimental Setup

We evaluate RAR on six public encrypted-traffic datasets that span the principal application and threat surfaces: mobile app traffic (NUDT-Mobile), DNS-over-HTTPS (CIC-DoHBrw), IoT attack traffic (CIC-IoT2023), TLS 1.3 application traffic (CSTNET-TLS1.3), Tor traffic (ISCX-Tor), and web application traffic (EBSNN-WebApp). All six are public, so reviewers can reproduce every number we report.

We compare RAR against two baseline groups, each isolating a different claim. *Group A* holds the encoder fixed at ET-BERT [4] and varies the class-incremental learning (CIL) strategy: Learning without Forgetting (LwF) [3], iCaRL [5], BiC [6], DER [7], Adapter [2],

Table 1: Six encrypted-traffic datasets used in evaluation.

Dataset	Domain	Public	#Cls	#Flows
NUDT-Mobile	Mobile app	Yes	300	496086
CIC-DoHBrw	DNS-over-HTTPS	Yes	6	49777
CIC-IoT2023	IoT attack	Yes	22	26413
CSTNET-TLS1.3	TLS 1.3 app	Yes	120	43243
ISCX-Tor	Tor	Yes	26	8518
EBSNN-WebApp	Web app	Yes	48	15336

Table 2: Baseline groups.

Group	Variable held fixed	Baselines
A	ET-BERT encoder	LwF, iCaRL, BiC, DER, Adapter, Fine-tune
B	Replay + head expansion	AppScanner, ETC-PS, FlowLens, CNN-based, CNN-emb, EBSNN, FS-Net, GraphDApp, MM4flow, ET-BERT

and Fine-tune (lower bound). *Group B* holds the CIL setup fixed at “old-sample replay plus classification-head expansion” and varies the traffic-classification backbone across ten models: AppScanner, ETC-PS, FlowLens, CNN-based, CNN-emb, EBSNN, FS-Net, GraphDApp, MM4flow, and ET-BERT [4]. *Group A* tests whether reconstruction-error routing beats classical CIL strategies on a fixed encoder; *Group B* tests whether the forgetting gap persists under stronger encoders.

Five metrics isolate the quantity each claim depends on. *old-before* measures accuracy on old classes after head expansion and *before* training on the new step. *old-after* measures accuracy on old classes after training on the new step. *new-after-train* measures accuracy on the new classes after the new step. *mix* reports combined accuracy on old plus new classes after the new step. *forgetting* equals *old-before* minus *old-after* and isolates training-induced forgetting, so the head-expansion drop alone does not contribute.

We report forgetting and mix as the headline pair: forgetting tests the contribution of C1 (sample-level knowledge ownership) directly, and mix captures the practical incremental accuracy that operators observe.

All experiments share one training and evaluation pipeline. We freeze the encoder at its pre-trained weights to isolate the routing mechanism from encoder learning; only the classifier heads, the reconstruction module, and the logit calibration layer receive gradients.

We use a two-step incremental protocol. A base step trains the initial head on the first half of classes; an incremental step expands the head, replays a fixed buffer of old samples, and trains both heads jointly under RAR’s soft routing.

4.2 Comparison with CIL Strategies

Against classical CIL strategies on a fixed ET-BERT encoder, RAR cuts forgetting by an average of 47.8% (range 35.5%–67.5%) across six datasets and raises mixed-class accuracy by up to 15.9% over the strongest CIL baseline (Adapter).

Replay-based methods (DER, iCaRL) outperform uniform distillation (LwF) under encrypted traffic, yet both still forget more than RAR on every dataset.

Fine-tune, which omits all anti-forgetting mechanisms, sets the lower bound. RAR reduces forgetting by 35.5%–67.5% over the strongest CIL baseline (Adapter) on every dataset.

Reconstruction-error routing replaces uniform distillation with sample-aware knowledge balancing, the structural change that closes the gap.

4.3 Backbone Study

On NUDT-Mobile, naive incremental training forgets across all nine traffic-classification backbones, which isolates the forgetting problem from the encoder choice.

Backbones with stronger pre-trained representations reach higher accuracy, yet forgetting spans 13.3%–58.4% across the nine naive backbones. Even the strongest (GraphDApp, 13.3%) forgets more than twice RAR’s 6.65%.

A stronger encoder does not buy back the old-class accuracy that head expansion costs. RAR alone drops forgetting below the naive floor (6.65% vs 13.3% for the strongest naive backbone, GraphDApp); routing is the operation that recovers it.

Table 4 reports the naive-incremental forgetting rate and mix accuracy for each of the nine backbones on NUDT-Mobile, with RAR shown for contrast.

4.4 Ablation Study

Each of RAR’s three design choices contributes a measurable share of the overall gain; removing any one reopens part of the forgetting gap. We report all three ablations on NUDT-Mobile; the same pattern holds on the other five datasets, which we omit for brevity.

Table 5 reports three ablation experiments on NUDT-Mobile, each removing one design choice from the full RAR model. Removing the reconstruction routing signal (C1) raises forgetting by 5.78 pp; reverting to a single head (C2) raises forgetting by 4.49 pp and drops mix by 9.77 pp; removing calibration (C3) raises forgetting by 3.41 pp. Each component contributes on a different axis: routing and dual-head separation dominate forgetting reduction, while calibration provides the final mix accuracy.

Reconstruction temperature τ (design knob). Figure 5 reports forgetting and mix on CSTNET-TLS1.3 as τ sweeps from 0.25 (near-deterministic head selection) to 4.0 (near-uniform fusion). Forgetting drops from 9.71% at $\tau=0.25$ to 7.78% at $\tau=4.0$, and mix rises from 91.03% to 92.06%. The curve flattens after $\tau=1.0$: the difference between $\tau=1.0$ and $\tau=4.0$ is under 0.2 pp on both metrics. RAR therefore exposes a tunable knob that converges quickly; the operator can fix $\tau=1.0$ (the fitted value) without further tuning.

4.5 Robustness

Over T incremental steps and a range of replay-buffer sizes, RAR maintains its lead over Adapter, the strongest CIL baseline.

We split the 40 new classes of CSTNET-TLS1.3 into $T = 5$ steps of 8 classes each. Fine-tune collapses from 75.6% to 22.0% mix accuracy after five steps. Adapter degrades from 91.3% to 74.8%. RAR drops only from 93.4% to 84.6%, widening its lead over Adapter from 2.1 pp at $T=1$ to 9.8 pp at $T=5$.

Table 3: Forgetting ($F\downarrow$, lower is better) and mixed-class accuracy ($M\uparrow$, higher is better). Mean over 3 seeds.

Method	Dataset											
	NUDT		DoH		IoT		TLS		Tor		Web	
	$F\downarrow$	$M\uparrow$	$F\downarrow$	$M\uparrow$	$F\downarrow$	$M\uparrow$	$F\downarrow$	$M\uparrow$	$F\downarrow$	$M\uparrow$	$F\downarrow$	$M\uparrow$
Fine-tune	95.40	33.71	91.58	36.74	88.52	11.17	90.99	46.52	86.25	41.31	86.01	43.19
LwF [3]	61.03	55.91	88.91	36.75	52.58	43.53	56.21	65.66	96.37	34.72	78.11	48.20
iCaRL [5]	25.15	81.66	33.87	73.25	12.39	79.47	28.87	80.83	73.60	69.90	37.77	73.97
BiC [6]	24.78	79.48	33.87	73.25	11.68	80.09	28.94	80.65	84.87	42.21	38.15	73.61
DER [7]	12.64	86.15	33.92	73.19	11.33	79.37	16.85	85.11	77.28	46.83	17.89	84.78
Adapter [2]	12.79	86.86	33.86	73.26	9.22	82.12	16.40	87.20	13.11	89.16	13.98	88.69
RAR (ours)	6.65	90.12	11.00	84.93	4.82	86.05	8.49	91.70	8.45	91.94	8.37	92.30

Table 4: Naive-incremental forgetting and mix (%) on NUDT-Mobile.

Backbone	Forgetting (%)	Mix (%)
AppScanner	58.37	49.30
ETC-PS	56.63	47.99
FlowLens	56.99	45.69
CNN-based	44.10	58.66
CNN-emb	41.27	63.89
EBSNN	30.81	73.37
FS-Net	23.35	74.70
GraphDApp	13.26	55.54
MM4flow	22.45	81.61
RAR (ours)	6.65	90.12

Table 5: Ablation results on NUDT-Mobile.

Variant	Forgetting (%)	Mix (%)
w/o routing	12.43	87.29
w/o dual-head	11.14	80.35
w/o calibration	10.06	87.82
RAR (full)	6.65	90.12

RAR’s forgetting rate rises only from 5.98% to 11.82% as the memory ratio drops from 0.05 to 0.005. Adapter degrades from 7.97% to 22.70% over the same range. At the smallest buffer (memory ratio 0.005), RAR’s forgetting (11.82%) stays below Adapter’s forgetting at twice the buffer size (11.70% at ratio 0.02).

Figure 6 reports the per-step accuracy and buffer sensitivity on CSTNET-TLS1.3.

5 Related Work

5.1 Class-Incremental Learning

The classical CIL literature splits into distillation-based, calibration-based, and replay-based families. Each family optimizes a sample-agnostic signal that cannot distinguish old-learning from new-learning samples.

Distillation-based methods (LwF [3], iCaRL [5]) match the soft outputs of the old model on every sample with equal weight. The

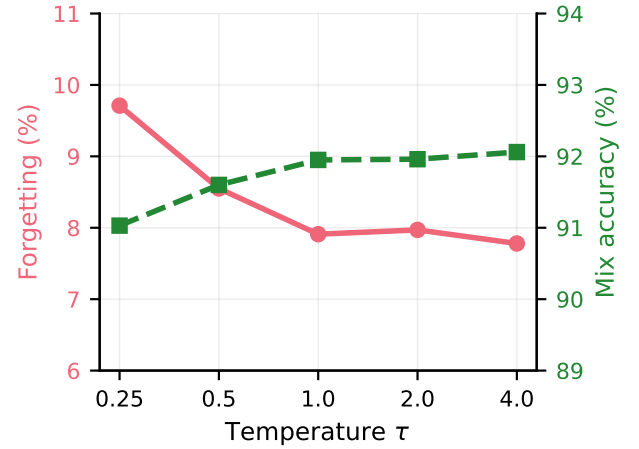


Figure 5: Forgetting and mix as a function of τ , on CSTNET-TLS1.3.

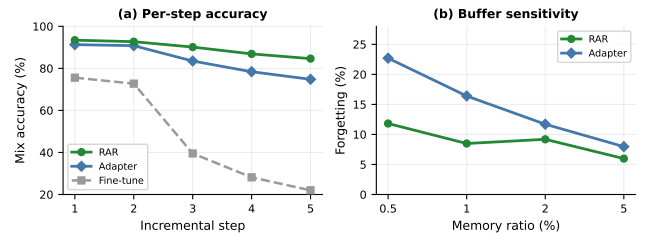


Figure 6: Robustness on CSTNET-TLS1.3: (a) per-step mix accuracy over $T = 5$ steps, and (b) forgetting as a function of memory ratio.

structural limitation is that the soft output carries class-boundary information from the old model and therefore biases the new model toward the old boundary regardless of whether the input is old or new.

Calibration-based methods (BiC [6]) correct the bias between old and new logits after training, but they apply a single scalar correction per class. Two samples with the same true class therefore

receive the same bias regardless of their reconstruction difficulty. Bias correction at the class level cannot recover the per-sample distinction.

Replay-based methods (DER [7]) store old samples or features and replay them during incremental training. The structural limitation is the buffer: a fixed buffer cannot represent the full old-class distribution, and the model's behavior on unstored samples inherits the buffer's under-representation.

RAR replaces the sample-agnostic signal with a per-sample old-versus-new probability derived from reconstruction error, so each sample contributes a sample-aware weight to the distillation, calibration, and replay terms.

5.2 Traffic Classification Backbones

Encrypted-traffic classification research has produced a series of strong backbones (ET-BERT [4], AppScanner, ETC-PS, FlowLens, CNN-based, CNN-emb, EBSNN, FS-Net, GraphDApp, MM4flow) that pre-train on packet-level features and reach high accuracy on static label sets. The structural limitation of this line of work is that it solves representation learning, not incremental learning: a stronger backbone still forgets old classes when its classification head expands to accommodate new ones. We quantify this in §4.3: naive incremental training forgets across all nine backbones on NUDT-Mobile, and RAR (on top of ET-BERT) breaks out of the band.

RAR's reconstruction-error routing operates on the frozen encoder's features and inherits the encoder's separability. The routing mechanism does not require the encoder to be incremental-aware; the per-sample knowledge ownership probability provides a general signal that can be incorporated into frozen-encoder CIL pipelines. Recent work on incremental representation learning, such as MEMENTO [1], adapts the backbone to the class stream rather than fixing it. In contrast, RAR studies the complementary routing problem, and the reconstruction-derived ownership signal could in principle be combined with representation-level updates.

6 Conclusion

RAR transforms per-sample reconstruction errors into knowledge ownership probabilities, enabling adaptive soft routing between old and new classification heads. By introducing reconstruction-aware routing, dual-head decoupling, and logit calibration, RAR effectively mitigates knowledge interference in encrypted traffic class-incremental learning. Extensive experiments on six encrypted-traffic datasets demonstrate that RAR consistently reduces forgetting while improving mixed-class recognition performance over existing CIL baselines.

Although effective, RAR still relies on limited replay samples to characterize old knowledge and may face challenges when old and new classes exhibit highly similar representations. Future work will explore more adaptive knowledge routing mechanisms and integrate reconstruction-aware routing with incremental representation learning to further enhance continual traffic classification.

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